PRIVATE VERSUS PUBLIC BLOCKCHAINS

Broadly, there are two types of entities that can own the hardware supporting blockchains: public and private. The difference between public and private blockchains is similar to that between the Internet and intranets. The Internet is a public resource. Anyone can tap into it; there's no gatekeeper. Intranets, on the other hand, are walled gardens used by companies or consortiums to transmit private information. Public blockchains are analogous to the Internet, whereas private blockchains are like intranets. While both are useful today, there's little debate that the Internet has created orders of magnitude more value than intranets. This is despite vociferous proclamations by incumbents in the 1980s and 1990s that the public Internet could never be trusted. History is on the side of public networks, and while history doesn't repeat, it does often rhyme.[4]

The important distinction boils down to how the entities get access to the network. Remember, a blockchain is created by a distributed system of computers that uses cryptography and a consensus process to keep the members of the community in sync. A blockchain is useless in isolation; one might as well use a centralized database. The community of computers building a blockchain can either be public or private, commonly referred to as permissionless or permissioned.

Public systems are ones like Bitcoin, where anyone with the right hardware and software can connect to the network and